

Data Analytics II: Empirical Research Methods - Exercise 1

Aleksandar Maric

Akando Räderer

Daniel Margolin

28.02.2026

Exercise 1a

```
data <- read.csv("Ex1.csv")
summary(data)
```

```
##      carat      color      clarity      prise
## Min.   :0.2000 Length:48718 Length:48718 Min.    : 326
## 1st Qu.:0.3800 Class :character Class :character 1st Qu.: 893
## Median :0.6000 Mode  :character Mode  :character Median : 2026
## Mean   :0.6968                      Mean   : 2892
## 3rd Qu.:1.0100                      3rd Qu.: 4406
## Max.   :3.1100                      Max.    :10000
##      cut
## Length:48718
## Class :character
## Mode  :character
##
##
##
```

Renaming the misspelled columns with

```
data <- data |> rename(color=c0lor)
data <- data |> rename(price=prise)
```

Removing an unnecessary information from the data set -> color

```
data <- data |> select(-color)
```

Now we check for missing values

```
skimr::skim(data)
```

Table 1: Data summary

Name	data
Number of rows	48718
Number of columns	4
Column type frequency:	
character	2
numeric	2
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
clarity	4878	0.9	2	4	0	8	0
cut	0	1.0	4	9	0	5	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
carat	0	1	0.70	0.36	0.2	0.38	0.6	1.01	3.11	
price	0	1	2892.25	2399.59	326.0	893.00	2026.0	4406.00	10000.00	

and we see, that in the column “clarity” there are almost 5k missing entries. Removing them finally gives us a clean dataset.

```
data <- data |> filter(!is.na(clarity))
```

Now we compute the averages and the standard deviations:

```
mean(data$carat)
```

```
## [1] 0.6964464
```

```
sd(data$carat)
```

```
## [1] 0.3571359
```

```
mean(data$price)
```

```
## [1] 2891.566
```

```
sd(data$price)
```

```
## [1] 2399.435
```

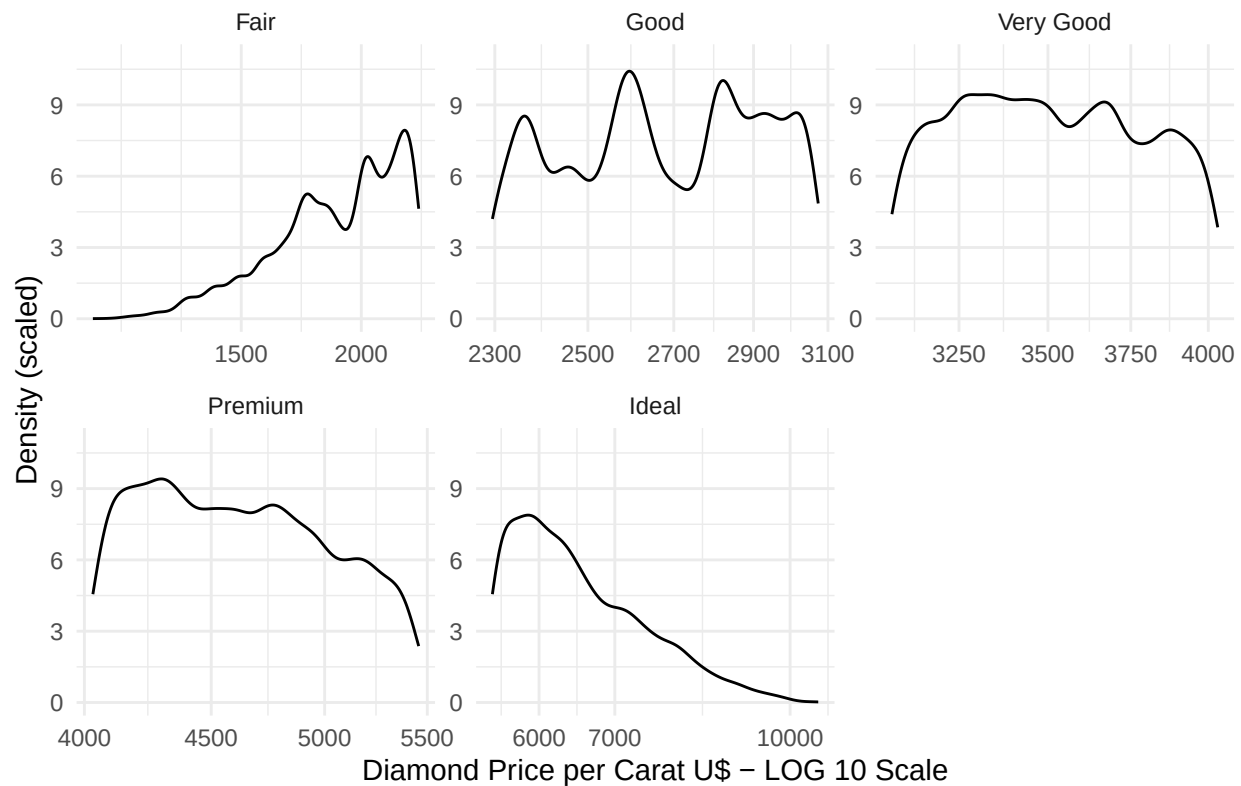
And finally produce a density plot to visualize the distribution of price per carat, while grouped by the cut.

```
data$cut <- factor(data$cut, levels = c("Fair", "Good", "Very Good", "Premium", "Ideal"), ordered = TRUE)
```

With `ggplot` we can plot the graphs. With `facet_wrap()` we can create plots for each category we just split the data into. The scale is also changed to logarithmic for better readability.

```
ggplot(data = data, aes(x=price/carat))+  
  geom_density()+  
  facet_wrap(~(cut), scale = "free")+  
  scale_x_log10()+  
  ylim(0, 11) +  
  ggtitle("Distribution of Diamond Price per Carat: grouped by cut.") +  
  xlab("Diamond Price per Carat U$ - LOG 10 Scale") +  
  ylab("Density (scaled)") +  
  theme_minimal()
```

Distribution of Diamond Price per Carat: grouped by cut.



From the chart, we can observe that the price per carat increases with cut quality. However, this result does not imply that cut quality increases price / carat causally. Further analysis would be needed to draw conclusions in this regard. You will learn the tools to conduct such analyses during the rest of the semester.